

TechCorp RAG Pipeline Documentation

Retrieval-Augmented Generation Architecture & Operations

Version: v1.8

Document: TC-RAG-004

TechCorp Internal

1. System Architecture

TechCorp's RAG pipeline is a multi-stage system that transforms raw documents into grounded, accurate LLM responses. The pipeline is built on a microservices architecture with each stage independently scalable.

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[Document Sources] --> [Ingestion Service] --> [Chunking Engine] --> [Embedding Service] --> [Vector DB (Qdrant)] | [User Query] --> [Query Processor] --> [Retriever] --> [Reranker] | [LLM (Claude / GPT-4o)] | [Response + Citations]
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2. Document Ingestion

Supported Formats	PDF, DOCX, PPTX, HTML, Markdown, TXT, CSV, JSON
Ingestion Method	REST API upload, S3 bucket watch (SQS trigger), webhook
Throughput	500 documents/minute at peak (auto-scaled ECS tasks)
De-duplication	SHA-256 hash of raw bytes — skip if already indexed
Metadata Extraction	Title, author, created_at, file_type, tenant_id, tags
Pre-processing	Remove headers/footers, normalize whitespace, fix encoding
Status Tracking	Async job queue (Celery + Redis); webhook on completion

3. Chunking Strategy

- Default strategy: Recursive Character Text Splitter — chunk size 512 tokens, overlap 64 tokens
- Semantic chunking (sentence-boundary aware) used for legal and medical documents
- Hierarchical chunking: parent chunk (1024 tokens) stored alongside child chunks for context expansion
- Tables and code blocks treated as atomic chunks — never split mid-structure
- Each chunk stores: content, chunk_index, parent_doc_id, token_count, page_number
- Minimum chunk size: 50 tokens — smaller fragments are merged with adjacent chunks

4. Embedding Model

Primary Model	text-embedding-3-large (OpenAI) — 3072 dimensions
Fallback Model	sentence-transformers/all-mpnet-base-v2 — self-hosted
Batch Size	128 chunks per API call — optimal cost/latency trade-off
Normalization	L2 normalized before storage
Multilingual	paraphrase-multilingual-mpnet-base-v2 for non-English docs
Re-embedding	Triggered when model version is upgraded (full re-index)
Cost Control	Token caching: embeddings cached 30 days by content hash

5. Vector Database

Engine	Qdrant 1.9 — self-hosted on EKS, 3-node cluster
Distance Metric	Cosine similarity
Collections	One collection per tenant (namespace isolation)
Payload Filters	tenant_id, doc_type, created_at, tags — MUST filter before search
Capacity	~50M vectors per node, 768 dims baseline
HNSW Settings	m=16, ef_construction=128, ef=64 for search
Backup	Qdrant snapshots to S3 daily; 7-day retention

6. Retrieval

- Hybrid search: dense vector search (70% weight) + BM25 sparse search (30% weight)
 - Top-K = 20 candidates retrieved initially, reduced to top-5 after reranking
 - Mandatory payload filter on tenant_id before every search — enforced at query layer
 - Query expansion: user query + HyDE (Hypothetical Document Embedding) for better recall
 - MMR (Maximal Marginal Relevance) applied to reduce redundant results
 - Sub-100ms P95 retrieval latency target (excluding network)
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7. Reranking

Model	cross-encoder/ms-marco-MiniLM-L-6-v2 (self-hosted)
Input	Query + top-20 retrieved chunks
Output	Re-scored list; top-5 selected for context window

Threshold	Chunks with rerank score < 0.4 are discarded
Latency Budget	< 150ms for reranking pass
Fallback	If reranker unavailable, use raw retrieval scores

8. LLM Integration

Primary Model	claude-sonnet-4 (Anthropic)
Fallback	gpt-4o (OpenAI) — activated if Anthropic API errors > 3% in 1 min
Temperature	0.2 for factual Q&A; 0.7 for summarization
Max Output	1,024 tokens (configurable per use case)
Streaming	Enabled for interactive chat; disabled for batch processing
Retry Policy	Exponential backoff — 3 retries, max 30s total

9. Evaluation

Metric	Tool	Target	Frequency
Faithfulness	RAGAS	> 0.85	Per release
Answer Relevancy	RAGAS	> 0.80	Per release
Context Recall	RAGAS	> 0.78	Per release
Latency P95	Grafana	< 2.5s	Real-time
Hallucination Rate	Custom eval	< 3%	Weekly
User Rating	In-product thumbs	4.2/5	Monthly

10. Feedback Loop

- User thumbs-up/down captured per response, stored in feedback_events table
- Negative feedback triggers human review within 24 hours via Slack alert
- Weekly feedback analysis report auto-generated and shared with AI team
- Top failure patterns used to create new evaluation test cases
- Model fine-tuning candidates identified from curated negative feedback samples
- A/B testing framework enables gradual rollout of prompt/model changes